

Sara Rosato

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Education

- PhD Candidate, LMU** Munich, German 09/2026 – Present
- **Topic:** Machine Learning applied to healthcare
 - MCML affiliated
- M.Sc. in Data Science and Engineering (Double Degree), Politecnico di Torino** Turin, Italy 10/2022 – 03/2025
- **Thesis:** Enhancing Anomaly Detection in Semiconductor Manufacturing Using Deep Learning (Qualcomm RF360)
 - Grade: 108/110
- M.Sc. in Intelligent Communication Systems (Double Degree), EURECOM** Biot, France 09/2023 – 03/2025
- **Relevant Coursework:** Machine Learning, Statistical Signal Processing, Project Management
 - **Scholarship:** Awarded ERASMUS+ grant (top 300 candidates selected)
- B.Sc. in Computer Engineering, Politecnico di Torino** Turin, Italy 09/2019 – 10/2022

Projects

- Galois Project – SIGMOD Publication** | Java, LLMs, Database Systems
- Developed a database system integrating LLMs as a storage layer for SQL queries.
 - Designed custom physical operators and query optimizations, improving result quality and efficiency.
 - Published at **SIGMOD**: "Logical and Physical Optimizations for SQL Query Execution over Large Language Models".
- Semantic Segmentation for Autonomous Driving** | Python, PyTorch, Federated Learning
- Benchmarked state-of-the-art semantic segmentation models under real-world conditions (lighting, weather).
 - Proposed *Clustered Averaged Batch Normalization*, improving performance in federated learning setups.
- Master's Thesis – Qualcomm RF360** | Deep Learning, Computer Vision
- Applied anomaly detection techniques on high-resolution wafer microscopy images.
 - Achieved 90% accuracy in defect detection, supporting semiconductor manufacturing reliability.
- Methane Emission Detection (SLB Research)** | Computer Vision, Lidar Imaging
- Developing ML models to analyze lidar images of methane plumes.
 - Contributing to sustainability by improving detection and quantification of fugitive emissions.

Experience

- Data Engineer, SLB Research Center** Cambridge, MA 03/2025 – 08/2026
- Designing machine learning models to analyze lidar camera images of methane plumes.
 - Contributing to sustainability by improving detection and quantification of fugitive emissions.
- Data Engineer Intern, Qualcomm RF360** Munich, Germany 08/2024 – 01/2025
- Optimized ML pipelines and data architecture for semiconductor sensor systems.
- Data Analyst, Politecnico di Torino (CALOS Department)** Turin, Italy 04/2024 – 09/2024
- Preprocessed and analyzed 100+ sensor streams, detecting anomalies and long-term operational trends.

Teaching & Mentoring

- Mathematics Tutor, GoStudent** Remote 11/2022 – 03/2025
- Tutored 20+ high school students, raising math grades (5→9) and physics grades (6→8).
 - Produced 15+ video tutorials on advanced math topics, reaching 200+ learners.
- WeAreHere Mentor, Politecnico di Torino** Turin, Italy 09/2022 – 10/2024
- Mentored 60+ female STEM students; coordinated 30+ mentors as lead organizer.
- Teaching Assistant – Linear Algebra, Politecnico di Torino** Turin, Italy 03/2022 – 07/2022
- Led tutorials and assessments; supported 80% of students in passing the exam on their first attempt.

Certifications & Skills

Certifications: Google Cloud Digital Leader, Udacity – AI Programming with Python, Foundations of Generative AI
Programming: Python, C, MATLAB, Java, LaTeX
ML/AI Tools: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy
Data Systems: SQL, MongoDB, Hadoop, NiFi, Tableau
Languages: Italian (Native), English (Proficient), French (Conversational)

Volunteering

- QWomen Employee Network, Qualcomm** 09/2024 – 01/2025
- Organized STEM Family Day to promote STEM education and inclusivity.
 - Coordinated the annual Christmas event to foster employee community engagement.
- BEST Torino (Board of European Students of Technology)** 09/2021 – 11/2022
- Organized international student events and engineering challenges for 200+ participants.